

Tutorial

Accelerated Real-Time C++ Audio: DSP Offloading and GPU Neural Inference

Victor Zappi

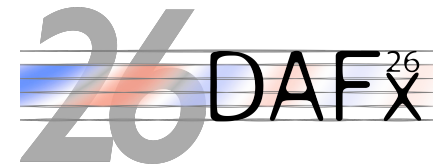
College of Arts Media and Design, Northeastern University

Get these slides here:



github.com/victorzappi/dafx26-accelerated-audio-tutorial

September 1, 2026



In a Nutshell

0. Intro/context/tools

1. CPU Audio

2. DSP Offloading

3. GPU Neural Audio Inference

**BENCHMARKING INTEGRATED GPU ACCELERATION OF REAL-TIME NEURAL
AUDIO INFERENCE ON SNAPDRAGON**

Avery Huang, Gautham Srinivasan

Northeastern University
Boston, USA

Akito van Troyer

Berklee College of Music
Boston, USA

Victor Zappi

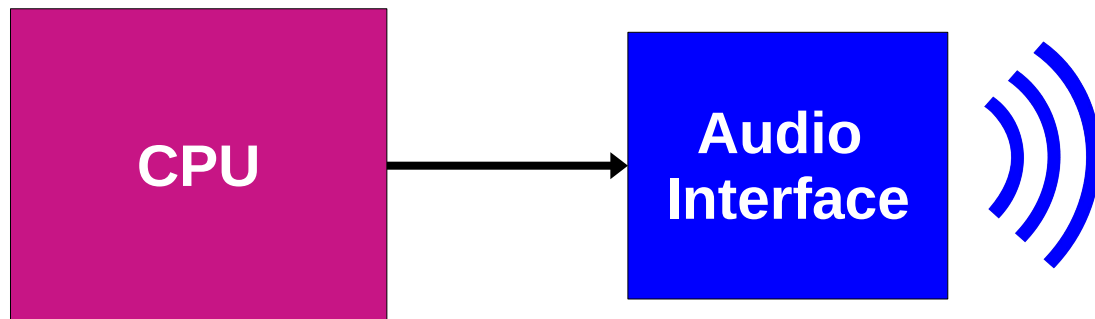
Northeastern University
Boston, USA

Paper Session 9, Friday 3pm

0. Intro/context/tools

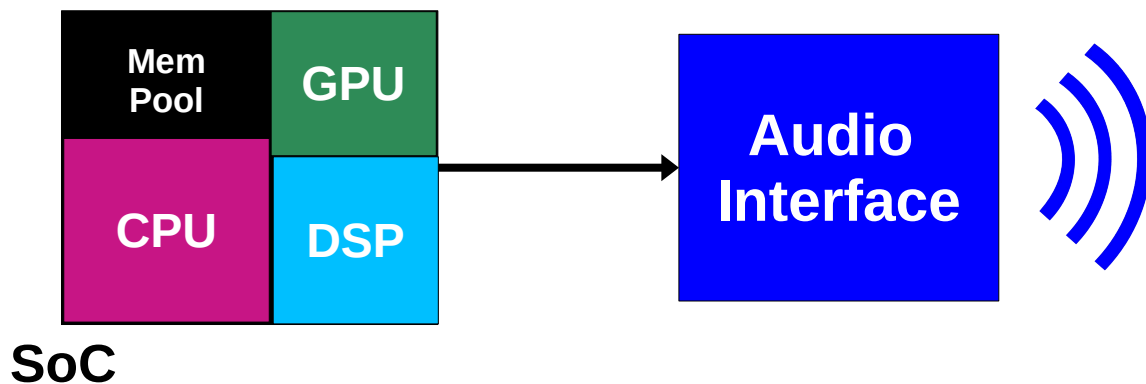
Standard Audio Workflow

- Typically, on general purpose computers the CPU is the processor where audio samples are synthesized/processed
- They are then sent to a *separate* device that turns the digital stream into an analog signal/sound (e.g., audio interface)



General Vs. Dedicated

- The CPU is designed to carry out all sorts of calculations, not just audio
- Audio doesn't have to run there though!
 - Nowadays, computers are often powered by **Systems on Chip** (SoCs)
 - SoCs pack *dedicated* processors alongside the general purpose CPU



Offloading Trade-off

- Offloading audio to dedicated processors can pay off in several ways:
 - Performance improvement
 - Algorithmic feasibility
 - CPU headroom

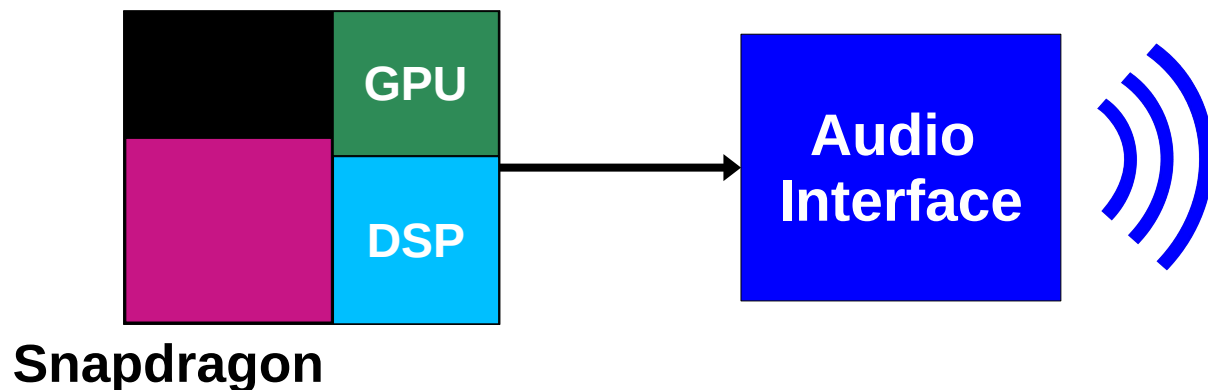
BUT

- It comes with some drawbacks:
 - Hardware and software dependent
 - Potentially complex setup
 - Constrains apply

Audio Acceleration on Snapdragon

- Today, we'll see how to use the dedicated processors of Qualcomm's Snapdragon family of SoCs as *audio accelerators*
- We'll cover:
 - audio offloading to the **Hexagon DSP**
 - neural audio inference on the **Adreno GPU**

Driven by custom C++ applications running from CPU!



Why Snapdragon?

- Snapdragons are found in a variety of computing devices, from general purpose computers to dedicated devices, including:
 - Laptops
 - Android Phones
 - Arduinos
 - Audio gears
- Qualcomm provides a convenient and efficient software ecosystem (frameworks, libraries, APIs) for running code on the dedicated processors

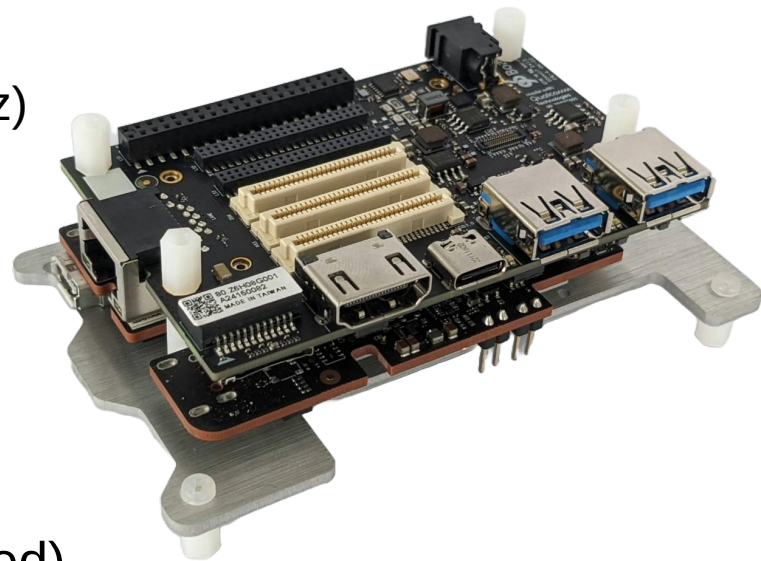


Board

- We'll use the RB3 Gen 2 board, a Snapdragon IoT development board
- The SoC – QCS6490 (Kodiak)
 - CPU: Kryo 670, 8 Arm Cortex cores (1.9–2.7 GHz)
 - DSP: Hexagon V68, 1980 MPPS
 - GPU: Adreno 643 @ 812 MHz

All recent Snapdragon SoCs are equipped with Hexagon+Adreno, so workflows are not specific to the RB3!

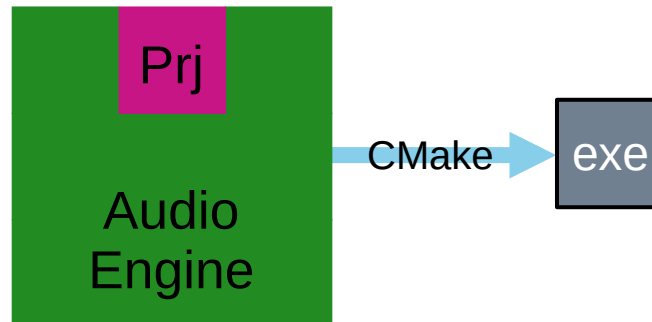
- Running Qualcomm Linux (Yocto/OpenEmbedded)
 - version 1.8, non-preemptive kernel



1. CPU Audio (on Snapdragon)

Audio Engine and Projects

- We'll build and run **audio applications in C++**, using an open-source **audio engine**:
 - github.com/victorzappi/ar-audioengine
- Each application is a **separate project** that gets built within the engine and runs on the board
- All the projects for this tutorial can be found in this repo:
 - github.com/victorzappi/dafx26-projects



A Minimal Project

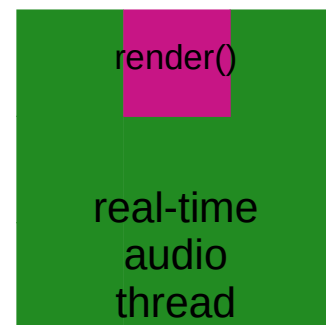
- Each project is composed of at least a *render.cpp* file, implementing a simple API:
 - *setup()*
 - *render()* ---> This is where samples are synthesized/processed and passed down the pipeline
 - *cleanup()*
- Let's inspect ***dafx26_projects/file_player...***

Audio Engine's Core

- At heart, the engine is **ordinary Linux audio code**
 - based on TinyALSA (lightweight ALSA userspace, standard on Android/embedded)
 - opens a PCM device (audio interface abstraction)
 - configures period size and period count (audio buffer), sets format and sample rate
 - real-time audio thread runs at highest priority, calls the project's *render()* function and pushes the samples to the PCM

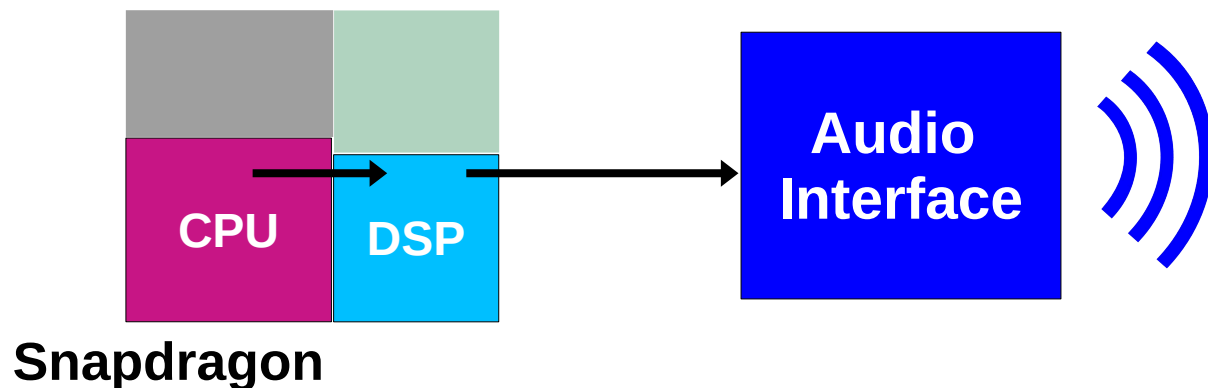
HOWEVER

- This is not enough for audio to work on Snapdragon!



DSP and Tunneling

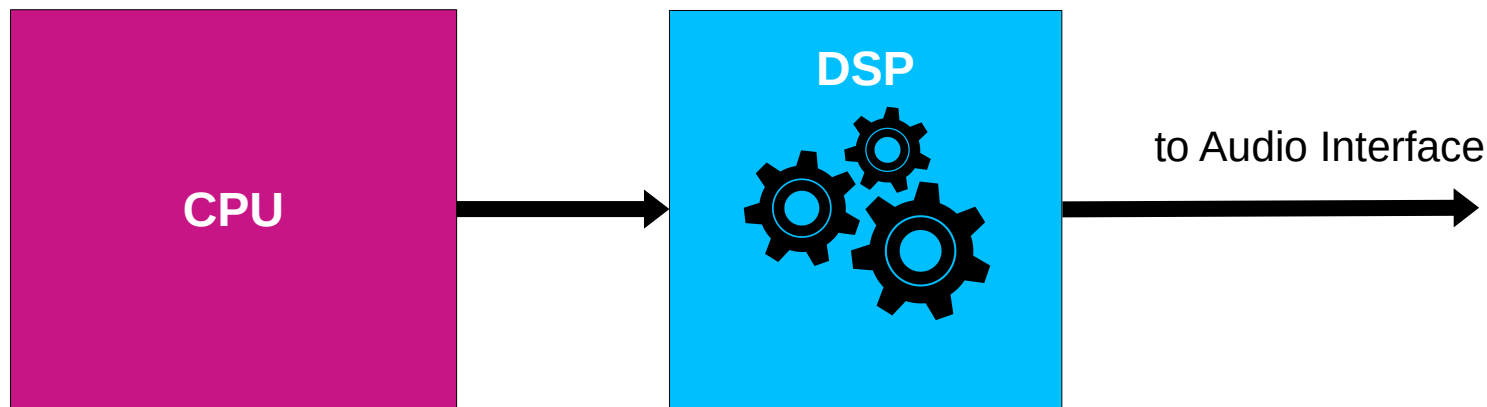
- On Snapdragon, the CPU cannot stream audio to the audio interface directly
- The stream **always passes through the DSP**
 - This is called *tunneling*



DSP Programming

- The DSP must be programmed to decide what happens to the samples as they travel through the DSP
 - This can include doing nothing, i.e., an actual pass-through!
- No DSP programming = cannot hear anything...

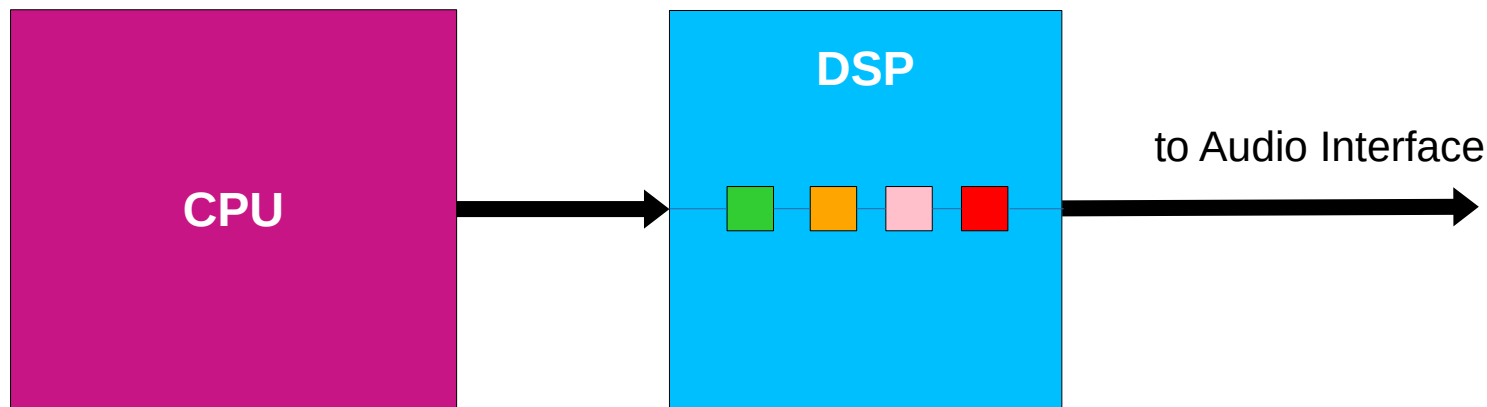
But how do we program the DSP?



AudioReach

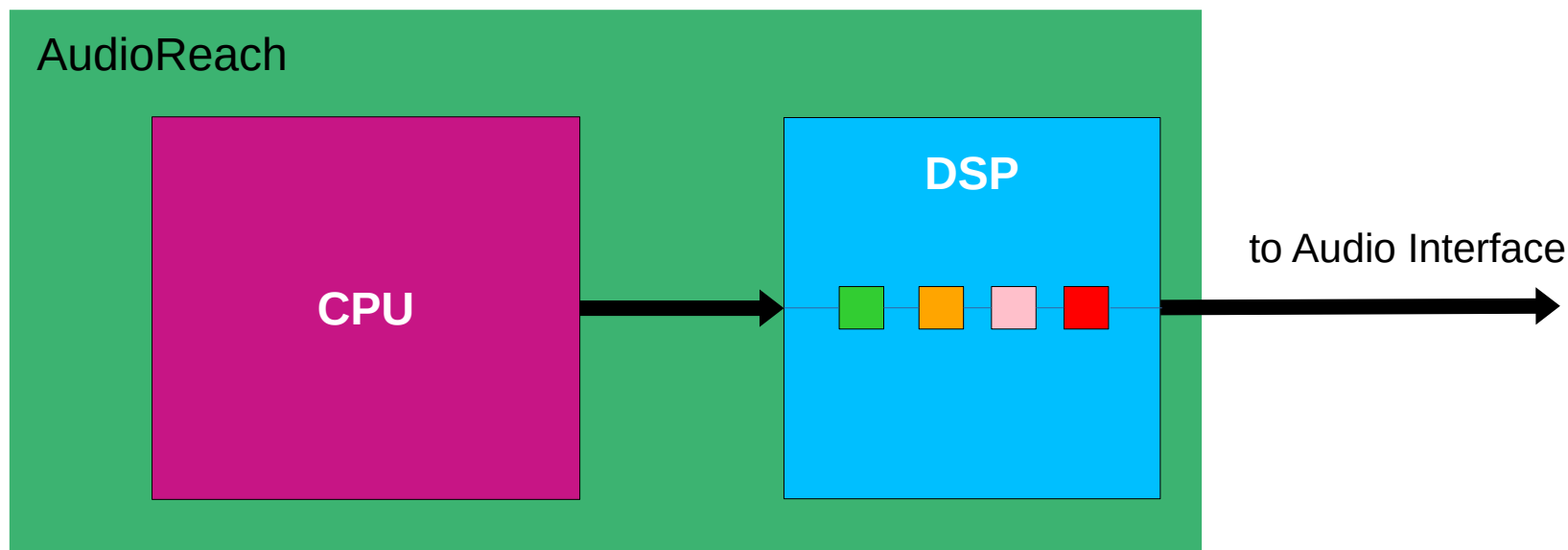
- In 2025 Qualcomm released *AudioReach*, a framework for programming the DSP via **graphs of processing modules**

*AudioReach is
open-source!*



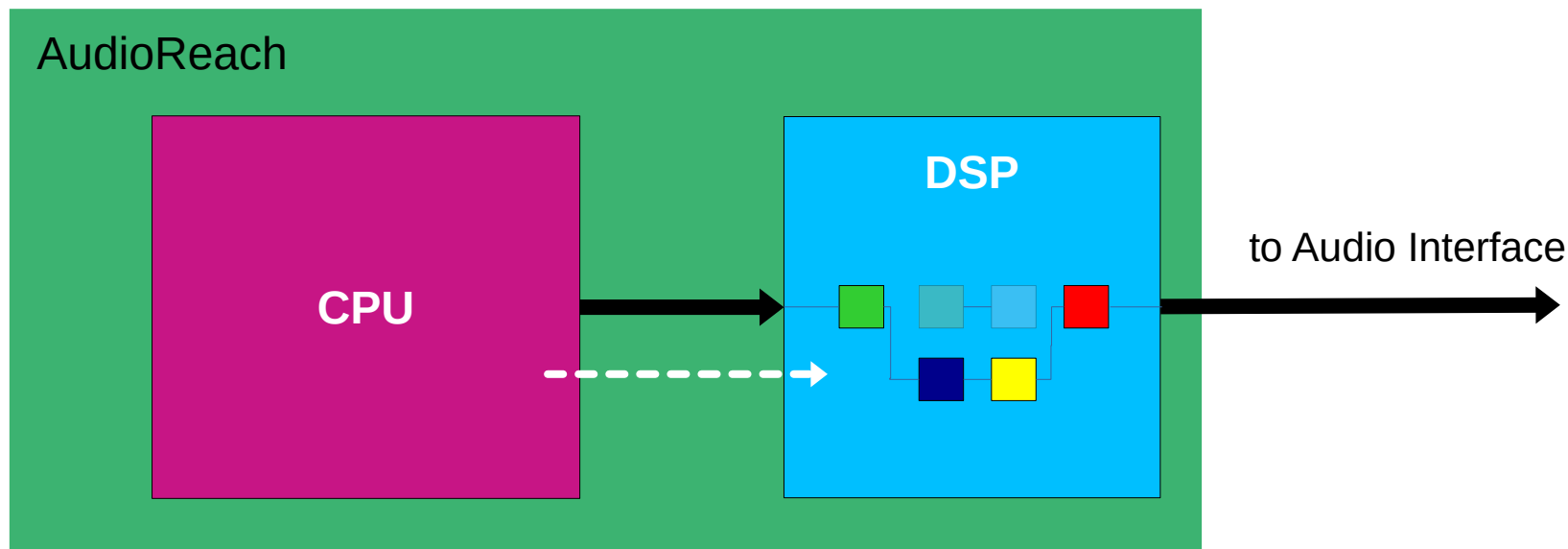
AudioReach

- In 2025 Qualcomm released *AudioReach*, a framework for programming the DSP via **graphs of processing modules**
- It also stretches to the CPU, allowing user code to configure the DSP at run-time



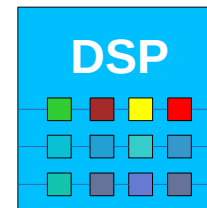
AudioReach

- In 2025 Qualcomm released *AudioReach*, a framework for programming the DSP via **graphs of processing modules**
- It also stretches to the CPU, allowing user code to configure the DSP at run-time
 - Our C++ code must tell AudioReach to load on the DSP the graph we want to use!



AudioReach DSP Programming

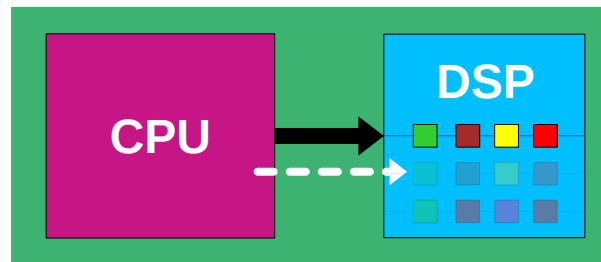
- The full AudioReach workflow is composed of two steps:
 - design a new graph (offline)



*All graphs
are stored
on device!*

AudioReach DSP Programming

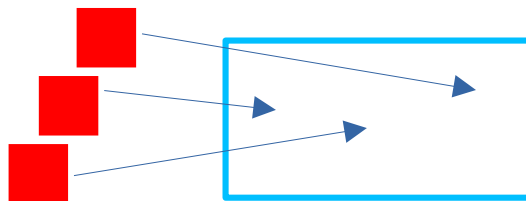
- The full AudioReach workflow is composed of two steps:
 - design a new graph (offline)
 - load that graph (run-time)



Let's see some details of this workflow!

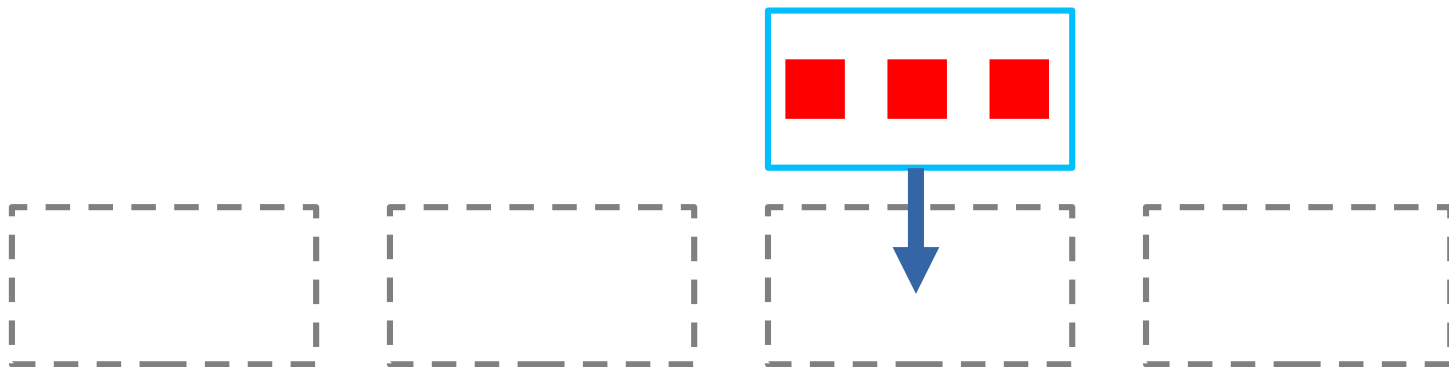
Graph Design (Simplified)

- AudioReach comes with *AudioReachCreator*, a graphical editor to create and edit graphs and including a library of modules
- To build a graph, first **modules** are grouped into **sub-graphs** (SGs)



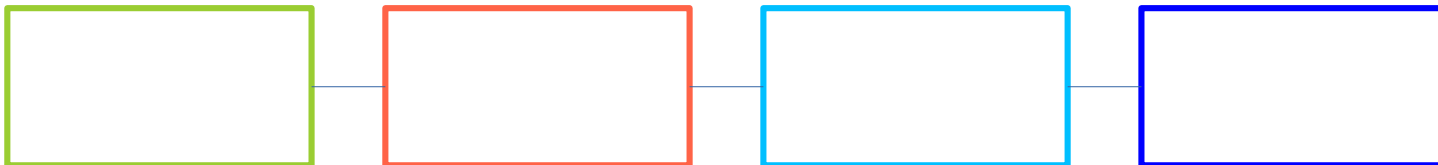
Graph Design (Simplified)

- AudioReach comes with *AudioReachCreator*, a graphical editor to create and edit graphs and including a library of modules
- To build a graph, first modules are grouped into **sub-graphs** (SGs)
- Then each sub-graph is placed in one of four possible slots within the graph



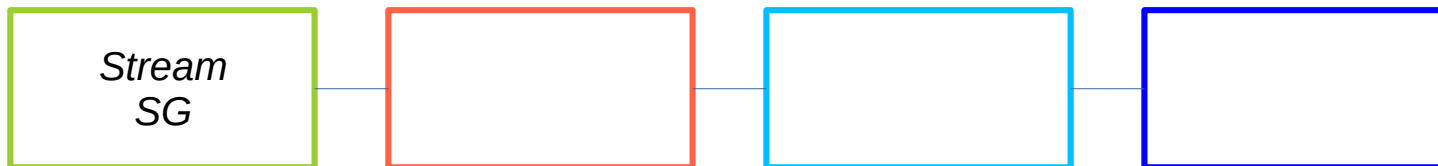
Graph Design (Simplified)

- AudioReach comes with *AudioReachCreator*, a graphical editor to create and edit graphs and including a library of modules
- To build a graph, first modules are grouped into **sub-graphs** (SGs)
- Then each sub-graph is placed in one of four possible slots within the graph
- Slots are chosen on the basis of what the sub-graphs' modules do:



Graph Design (Simplified)

- AudioReach comes with *AudioReachCreator*, a graphical editor to create and edit graphs and including a library of modules
- To build a graph, first modules are grouped into **sub-graphs** (SGs)
- Then each sub-graph is placed in one of four possible slots within the graph
- Slots are chosen on the basis of what the sub-graphs' modules do:

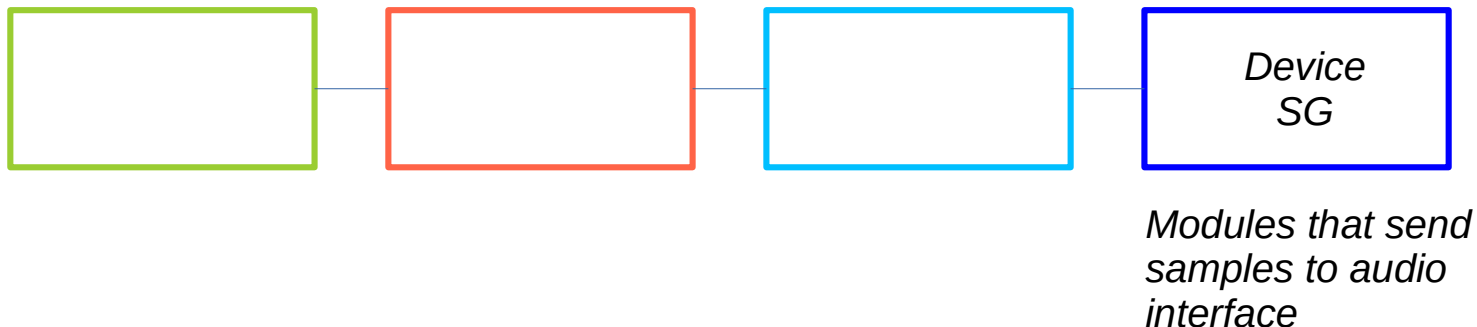


Modules that:

- *receive samples from CPU*
- *control volume*

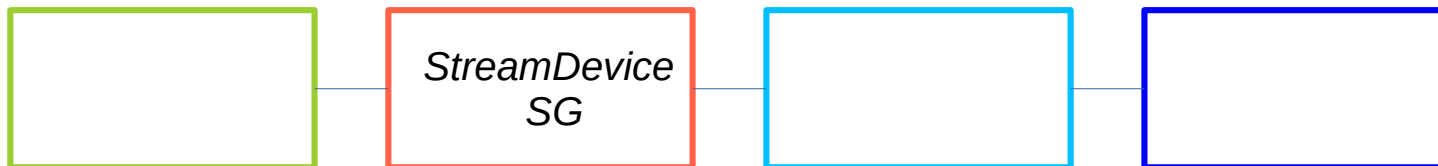
Graph Design (Simplified)

- AudioReach comes with *AudioReachCreator*, a graphical editor to create and edit graphs and including a library of modules
- To build a graph, first modules are grouped into **sub-graphs** (SGs)
- Then each sub-graph is placed in one of four possible slots within the graph
- Slots are chosen on the basis of what the sub-graphs' modules do:



Graph Design (Simplified)

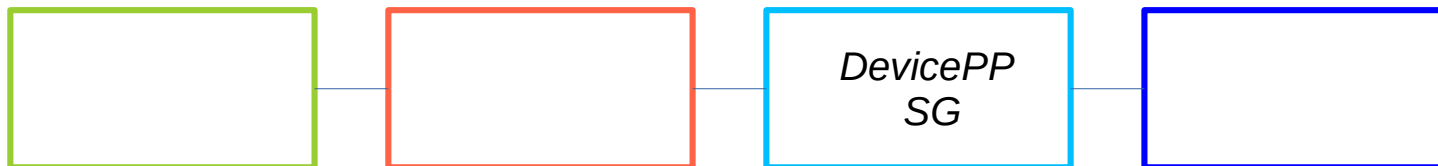
- AudioReach comes with *AudioReachCreator*, a graphical editor to create and edit graphs and including a library of modules
- To build a graph, first modules are grouped into **sub-graphs** (SGs)
- Then each sub-graph is placed in one of four possible slots within the graph
- Slots are chosen on the basis of what the sub-graphs' modules do:



*Modules that do
format and sample
rate conversion*

Graph Design (Simplified)

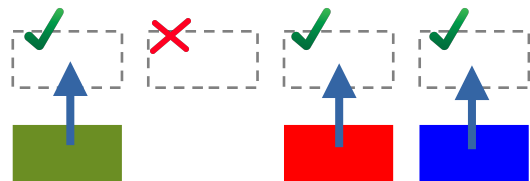
- AudioReach comes with *AudioReachCreator*, a graphical editor to create and edit graphs and including a library of modules
- To build a graph, first modules are grouped into **sub-graphs** (SGs)
- Then each sub-graph is placed in one of four possible slots within the graph
- Slots are chosen on the basis of what the sub-graphs' modules do:



*Modules that carry out
any processing before
output to device*

Graph Loading (Simplified)

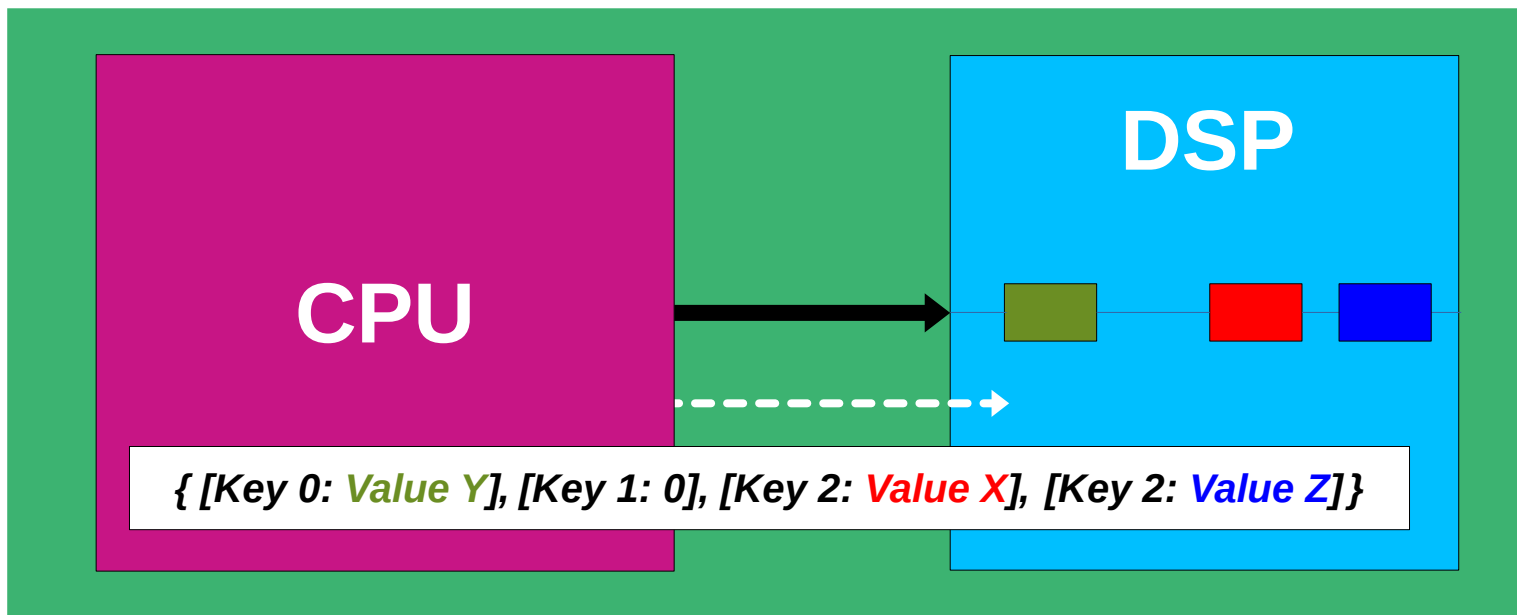
- Once the graph is built, it's identified by:
 - the list of slots that are filled (out of four)
 - what fills them (sub-graphs and their modules)
- AudioReach uses a **key-value system** to manage the identification process:
 - Each slot is associated with a fixed **key**
 - Each sub-graph is associated with a chosen numerical **value**



Value 0 means 'no sub-graph/empty slot'

Graph Key Vector

- CPU applications must assemble **a list of key–value** pairs to describe the graph that needs to be loaded on the DSP
- This list, called the *Graph Key Vector*, is sent to AudioReach at run time to identify the graph and load it!



Audio Engine Default Graph

- The audio engine we are using sends via the AudioReach API the following default Graph Key Vector:

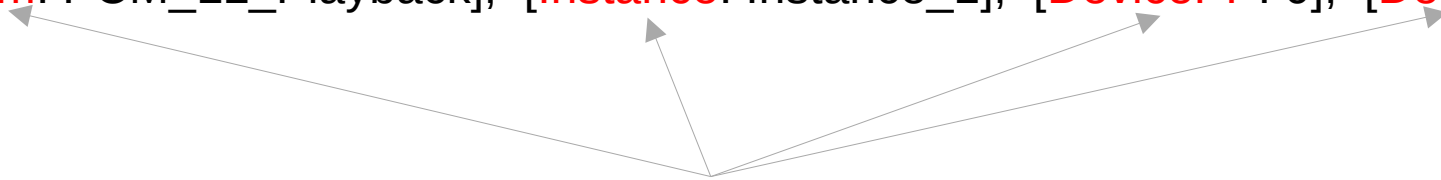
```
{ [Stream: PCM_LL_Playback], [Instance: Instance_1], [DevicePP: 0], [Device: Speaker] }
```

Audio Engine Default Graph

- The audio engine we are using sends via the AudioReach API the following default Graph Key Vector:

{ [**Stream**: PCM_LL_Playback], [**Instance**: Instance_1], [**DevicePP**: 0], [**Device**: Speaker] }

Fixed **keys**

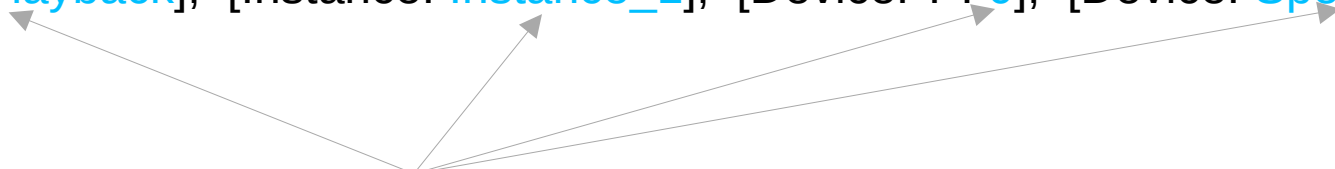


Audio Engine Default Graph

- The audio engine we are using sends via the AudioReach API the following default Graph Key Vector:

{ [Stream: **PCM_LL_Playback**], [Instance: **Instance_1**], [DevicePP: **0**], [Device: **Speaker**] }

Values (as macros)



Audio Engine Default Graph

- The audio engine we are using sends via the AudioReach API the following default Graph Key Vector:

```
{ [Stream: PCM_LL_Playback], [Instance: Instance_1], [DevicePP: 0], [Device: Speaker] }
```

- This means that all projects will load the same default graph on the DSP
- Let's open AudioReachCreator and have a look at this graph...

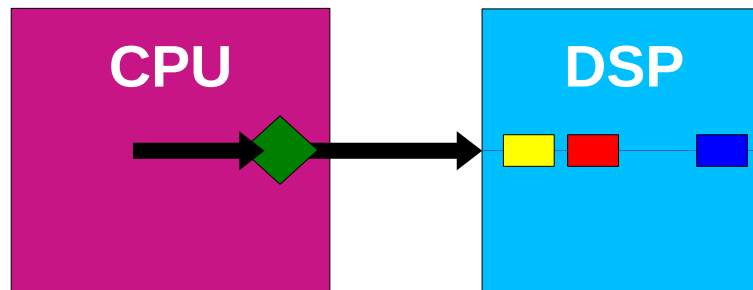
2. DSP Offloading

Why DSP?

- DSPs are processors designed to process streams of digital signals
- Most common operations in digital signal processing: filters, convolution, mixing, transforms
 - All composed of sums of products
 - All working on buffers
- Hence DSPs:
 - support **MAC** (multiply–accumulate) as a single instruction and are equipped with a few MAC units working in **parallel**
 - handle **circular buffers in hardware**—no index arithmetic needed
 - are designed for **fixed-point** arithmetic (integer math) with **saturation** (overflow clips instead of wrapping!)

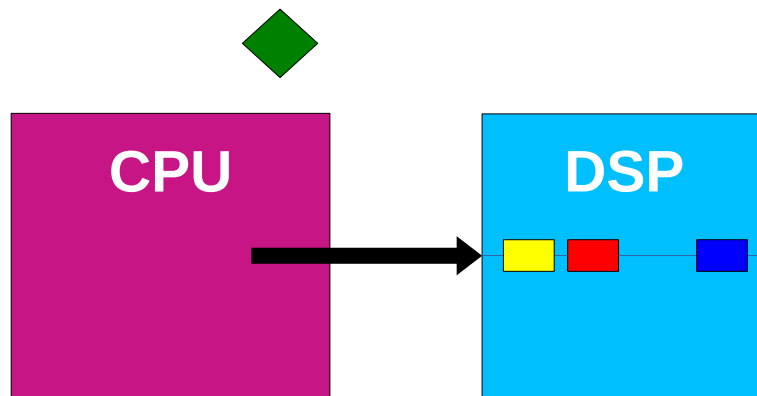
Offloading on Snapdragon

- Let's assume we designed a CPU audio application that runs some **processing** on the incoming samples (e.g., an audio effect)



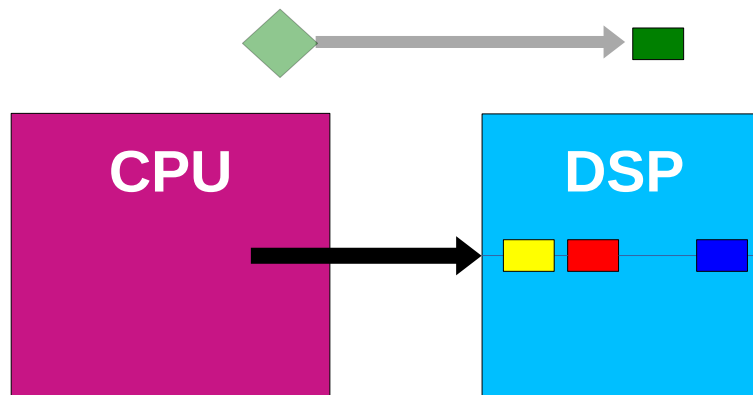
Offloading on Snapdragon

- Let's assume we designed a CPU audio application that runs some **processing** on the incoming samples (e.g., an audio effect)
- To offload that processing to the DSP, we remove it from the CPU code...



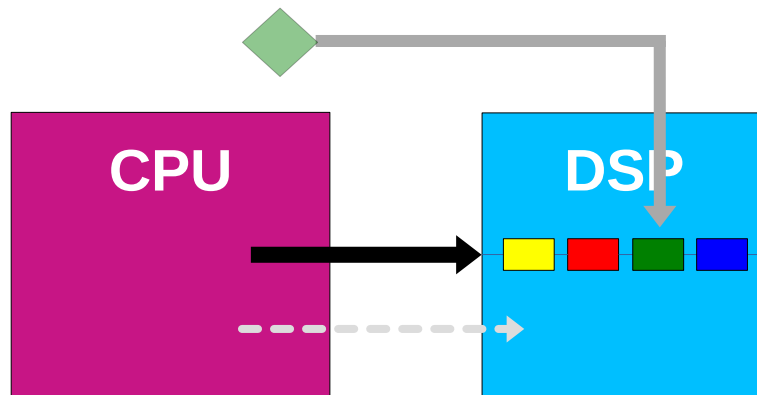
Offloading on Snapdragon

- Let's assume we designed a CPU audio application that runs some **processing** on the incoming samples (e.g., an audio effect)
- To offload that processing to the DSP, we remove it from the CPU code...
- ...and re-implement the same algorithm as a **dedicated sub-graph**, composed of one or more modules



Offloading on Snapdragon

- Let's assume we designed a CPU audio application that runs some **processing** on the incoming samples (e.g., an audio effect)
- To offload that processing to the DSP, we remove it from the CPU code...
- ...and re-implement the same algorithm as a **dedicated sub-graph**, composed of one or more modules
- Finally we create a new graph with it and load it at run time!



Offloading Example

- Let's have a look at the project ***projects/file_player_reverb...***

Free Lunch?

- Offloading relieved the CPU load... but is there any **hidden cost**?

Yes /:

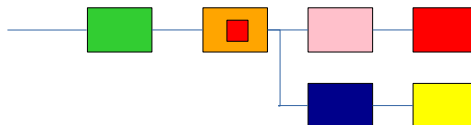
- When we add a new sub-graph, we increase the **latency** of the overall DSP chain!
 - A new sub-graph means a longer graph with more modules
 - Each module carries out some processing... and it needs time to do that!
 - AudioReach allocates a new **buffer** for the sub-graph, to give all its modules time to process the samples within real-time deadlines
- This is a real constraint of DSP offloading!
- Our new sub-graph is set to the lowest latency setting, i.e., 1 ms

Improving Performance

- Luckily, there are two ways around this added latency!
 - Graph design (AudioReachCreator)
 - Real-time audio parameters (CPU/run time)

Improving Performance – Graph Design

- New modules can be added inside *existing* sub-graphs
 - Same graph length often means same latency!
- Some rules apply though:
 - The DSP may not run fast enough to run all modules in real time (choking the DSP to relieve the CPU...)
 - Some specialized modules may still request additional computational time, increasing overall latency even if the number of sub-graphs remains the same
 - Sub-graphs can be shared across graphs! Editing one with a new module may impact other graphs that are designed for different use cases



AudioReachCreator is a sophisticated tool—graph design takes practice

Improving Performance – Audio Params

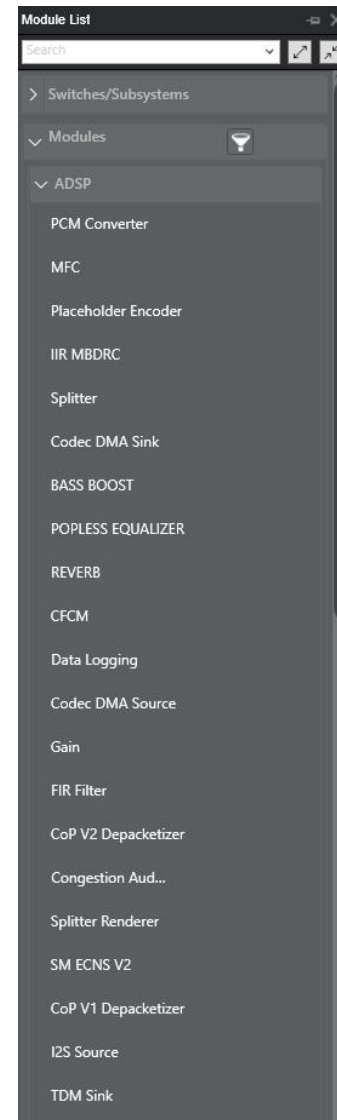
- When latency increases on the DSP side, it may be possible to reduce it on the CPU side!
- Offloading creates CPU overhead
- This may allow to run the CPU application with a much smaller audio buffer
 - Smaller audio buffer = lower latency!

Improving Our App's Performance

- The CPU app could run with a *period size* of no less than 96 frames
 - 96 frames @ 48 kHz → **4 ms CPU playback latency**
- With a ~9% CPU/core usage, our offloaded application's period size can be easily set to the minimum value supported by the hardware, i.e., 48 frames
 - Let's test it...
- Jumping from 96 to 48 frames means cutting down CPU latency in half!
 - **From 4 ms to 2 ms!**
- We not only canceled the **+1 ms** latency penalty of the DSP offload...
- ...but also improved the latency of the original application!
 - **Net -1 ms**

Custom DSP Modules

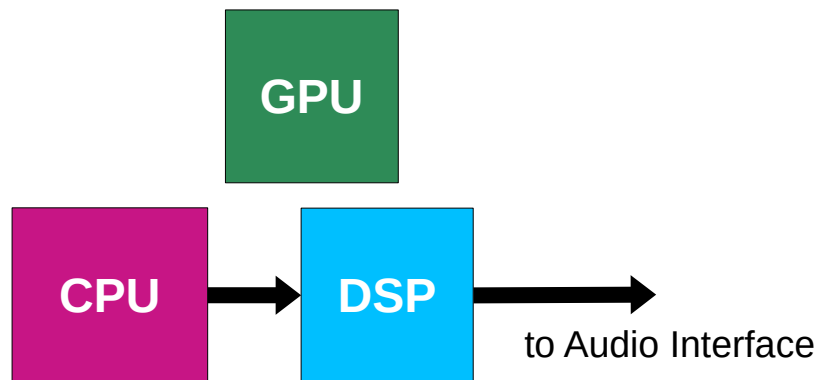
- AudioReachCreator includes a long list of modules that can be used and customized to design graphs
- But what if we want to offload an algorithm that is not implemented by any existing module?
 - You can build your own!
- AudioReach comes with *CAPI* (Common Audio Processor Interface), an API to write your own DSP modules in C



3. GPU Neural Audio Inference

GPU and Audio

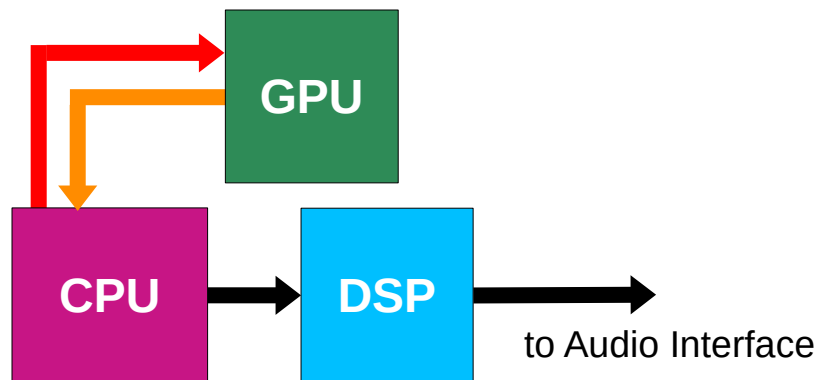
- On Snapdragon, audio always travels through the DSP to reach the audio interface, hence DSP offloading simply boils down to using a new/dedicated graph
- In contrast, the GPU is not a default part of the audio pipeline (on Snapdragon and anywhere else)!
 - Designed to process graphics
 - But flexible enough to carry out other tasks ('GPGPU'), including audio



GPU Audio Pipeline

- To use the GPU as an audio accelerator, we need to:
 - bring the **audio data to the GPU**, where it's processed
 - then bring the **result back to the CPU**
- From the CPU processed samples are again streamed into the standard DSP → interface pipeline

Issue 1: moving data can be costly!



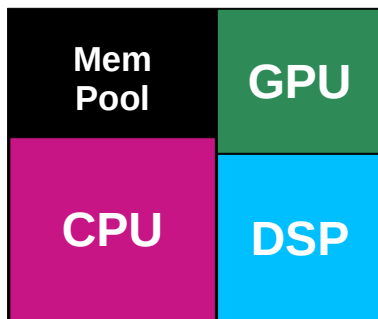
GPU Parallel Processing

- GPUs pack *thousands* of small cores (arithmetic and logic units), all running the same operation on different data at once
- Hence they require **kernels**, i.e., programs that execute simultaneously across as many cores as possible
 - Not a loop walking through data!
- If we want to run audio processing on a GPU, we need to implement our algorithm as a kernel

*Issue 2: writing parallelizable kernels is not simple—
especially neural models!*

Shared Memory

- On Snapdragon, the Adreno GPU shares a single memory pool with the rest of the silicon
 - Unified memory architecture
- This means that moving data to and from the GPU is less expensive than in other systems:
 - discrete GPUs (transfer across a bus to dedicated VRAM)
 - most integrated GPUs (same DRAM, but still a copy between separate address spaces)



*Issue 1:
largely solved!*

Qualcomm AI Runtime SDK

- Then, Qualcomm provides QAIRT (Qualcomm AI Runtime), an SDK for deploying neural models on Qualcomm hardware, including Snapdragon
- It contains:
 - tools to convert trained models into **optimized on-device representations** that are backend-independent (*backend* = target processor where the model will run)
 - **QNN**, a low-level API and the run-time libraries that load a converted model and execute it on a **chosen backend**
 - Including the GPU!

*Issue 2:
largely solved too!*

QAIRT Pipeline

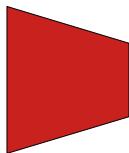
- We start from a **model** pre-trained in one of the main frameworks

 PyTorch

 TensorFlow

 ONNX

Model



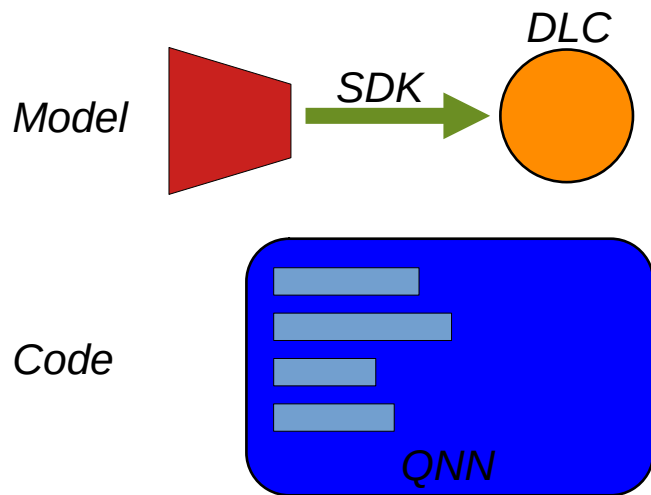
QAIRT Pipeline

- We start from a model pre-trained in one of the main frameworks
- Then with the **SDK** conversion tool we generate a **DLC** (Deep Learning Container)
 - i.e., the backend-independent format that is consumed at run time



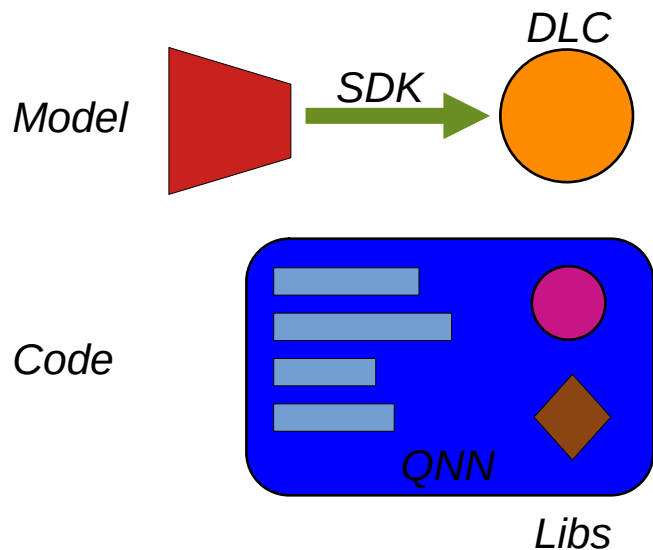
QAIRT Pipeline

- Our **C++ code** has to load two dynamic libs, accessible via the **QNN API**:



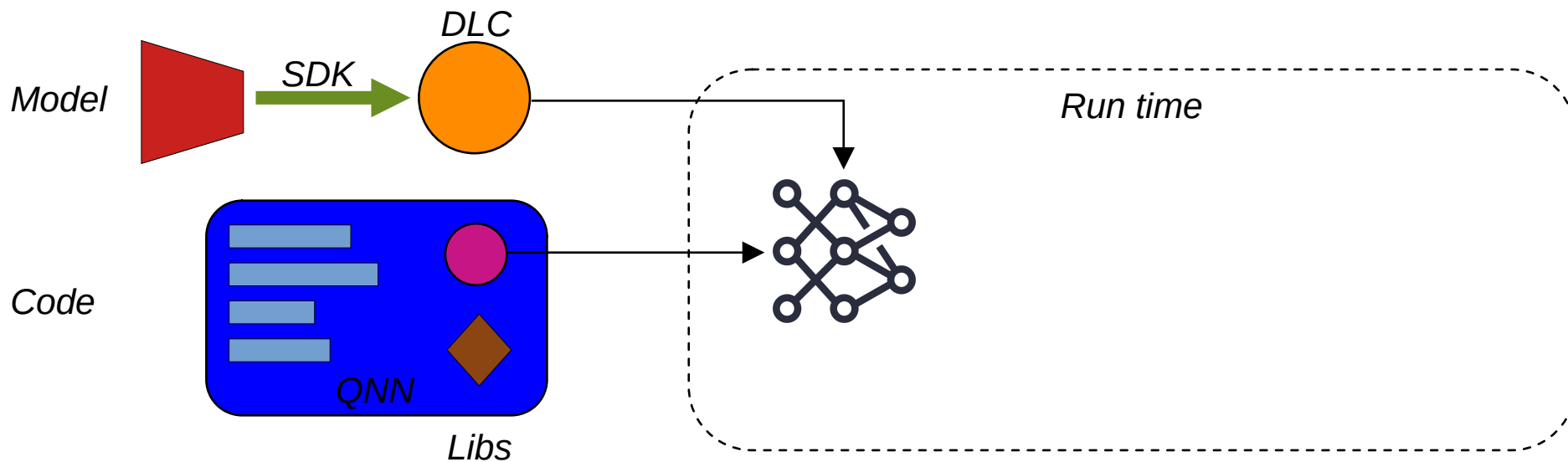
QAIRT Pipeline

- Our C++ code has to load two dynamic libs, accessible via the QNN API:
 - the **system library** (*libQnnSystem.so*)
 - a **backend library** (e.g., *libQnnCpu.so*, *libQnnGpu.so*)



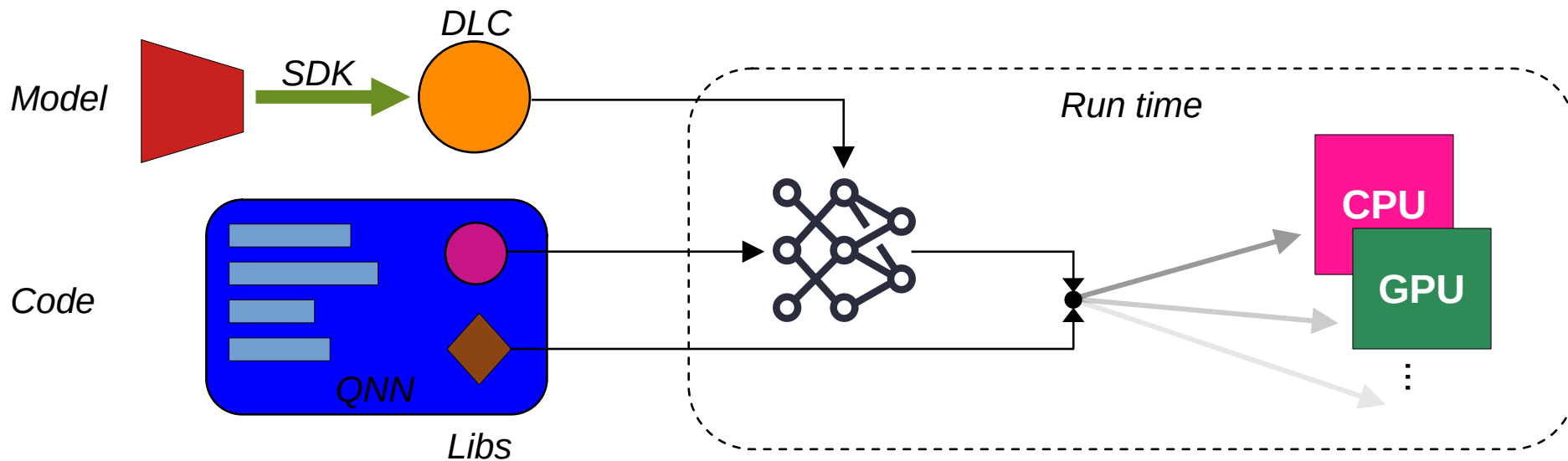
QAIRT Pipeline

- At run-time, the libs carry out different roles:
 - the system library parses the DLC and composes the model's **graph**



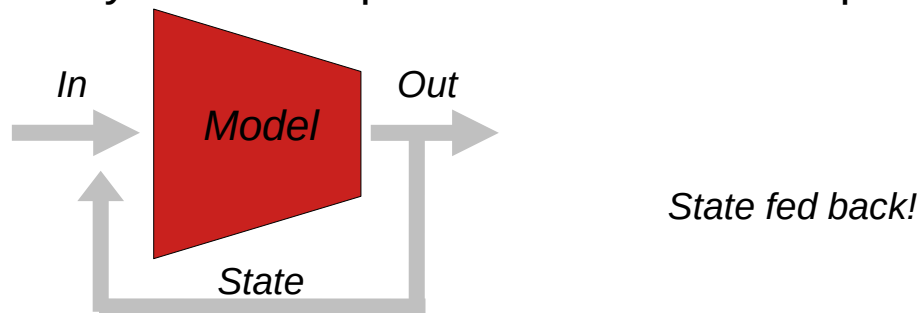
QAIRT Pipeline

- At run-time, the libs carry out different roles:
 - the system library parses the DLC and composes the model's graph
 - the chosen backend library finalizes that graph for its target backend, allocates the tensors and executes it



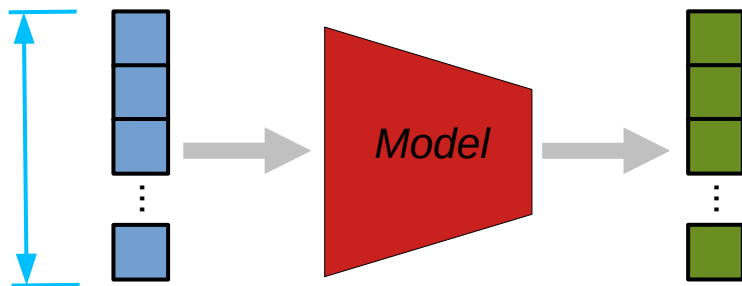
Neural Kernels

- When the target is the GPU, the backend maps each layer of the graph onto a pre-written parallel kernel
 - Kernels are designed in OpenCL and tuned for Adreno!
- This mapping is automatic (via QNN) but has a **caveat**...
 - All QNN graphs are **stateless**, i.e., nothing persists between inference calls
- This means that *stateful* Python models (e.g., those with hidden states, streaming caches) need to be modified **before conversion**
 - The model must pass any state as input and return it as output at every call



Neural Audio Kernels

- When running neural audio models, we have one inference call per audio callback (*render()* function)
- The model receives and processes a **buffer of samples** (neural processing) OR synthesizes a buffer of samples from scratch (neural synthesis)
- Each layer's kernel runs across all the samples simultaneously*
- Each inference returns a new buffer of samples
 - The **buffer size** determines how much **parallelism** the GPU gets!



Neural Audio Inference Example

- Let's have a look at the project ***projects/brave...***

Some Notes on GPU Acceleration

“Each layer's kernel runs across all the samples simultaneously”*

- *Actually, not all layers can parallelize across time!
 - For example, in recurrent layers sample $n+1$ depends on sample n , so the kernel has to run through the buffer sequentially, destroying the GPU speed-up
- In those that can run in parallel, the speed-up depends on the buffer size, but the buffer size also heavily impacts the overall latency of the application!
 - Small models may start showing meaningful speed-ups at buffer sizes that are already too large for interactive audio
- In the above cases, it may be better to stay on the CPU or to choose the DSP as the target backend
 - QNN has a dedicated libs for the Hexagon too, though models need to be quantized to *int16* or *int8*

Offloading $\stackrel{?}{=}$ Acceleration

- In our application, inference got offloaded to the GPU where it was accelerated
- Yet, in general **offloading and acceleration are not the same concept**
- In our reverb example, the algorithm was offloaded to the DSP, but we cannot claim it was accelerated
 - DSP is probably not faster here: 4 MACs/cycle at a low clock, against a 1.9 GHz Cortex-A55 with NEON
 - The win was CPU headroom, power and latency—not necessarily speed
- That said, the DSP can be a real accelerator!
 - Via optimized implementations (CAPI is an approach)